Math 401—Calculus II ***Exam 2*** ***Sp 2021***

Name_____ ***Date_____***

- *You may use a scientific calculator or a graphing calculator with capabilities such as a TI 84.* ***TI89 or TI-nspire models are not allowed.***
- *For full credit, fractional answers must be reduced and decimal answers must be rounded correctly.*
- *You may use scratch paper (mine only) but partial credit is only awarded to work that is partially correct on the exam.*

Level:

Question 1 …….. (12 points)

C6: INTEGRATION BY PARTS

A) $\int x^2 \cos(3x) dx$

B) $\int x^4 lnx dx$

Question 2 …… (12 points)

C7: INTEGRATION OF TRIGONOMETRIC FUNCTIONS

A) $\int sin^3x \cdot cos^2x dx$

B) $\int tan^4x dx$

Question 3 …… (12 points)

C8: INTEGRATION BY TRIGONOMETRIC SUBSTITUTION

A) Give the appropriate trigonometric substitution for the following integrals**. Don't evaluate** the Integral. Write out the substitution that would be used, including "dx".

i) $\int \sqrt{36 - x^2}\,dx$ (2 points)

ii) $\int \frac{dx}{x\sqrt{x^2-64}}$ (2 points)

iii) $\int \frac{x}{(25+4x^2)^{\frac{3}{2}}}\,dx$ (2 points)

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B) Use an appropriate trigonometric substitution to solve the integral:

$\int \frac{dx}{(x^2+9)^{\frac{3}{2}}}$ (6 points)

Question 4 …… (9 points)

C9: PARTIAL FRACTIONS

A) Write out the form of the partial fraction decomposition of the function. **Do not determine the numerical values of the coefficients.**

$$f(x) = \frac{x + 10}{(x^2 - 1)(x^2 + 9)}$$ (3 Points)

B) Use partial fractions to integrate: $\int \frac{x^2 + 2x - 1}{x^3 - x}\,dx$ (6 points)

Question 5 …… (12 points)
C10: INTEGRATION TECHNIQUES

You may use standard methods of integration for the following integrals:

A) $\int \frac{\sin 2q}{\sqrt{5 - \sin q}}\, dq$ (6 points)

B) $\int \sqrt{x} \cdot e^{\sqrt{x}}\, dx$ (6points)

Question 6 …… (6 points)
C11: IMPROPER INTEGRALS
Determine whether the integral converges or diverges:

$\int_2^{\infty} e^{-3x} dx$

Question 7 …… (6 points)
C5: Work in Lifting Fluids

The vertical ends of a watering trough are semicircles of radius 3 meters. If the trough is 6 meters long, find the work done in emptying the trough by pumping the water to the top of the trough is filled to within one meter from the top. The density of water is 1000kg/m^3

Some Useful formulas and tables of integration:

Sum and Difference Formulas

$$\sin(u \pm v) = \sin u \cos v \pm \sin v \cos u$$
$$\cos(u \pm v) = \cos u \cos v \mp \sin u \sin v$$
$$\tan(u \pm v) = \frac{\tan u \pm \tan v}{1 \mp \tan u \tan v}$$

Double Angle Formulas

$$\sin(2u) = 2\sin u \cos u$$
$$\cos(2u) = \cos^2 u - \sin^2 u$$
$$\tan(2u) = \frac{2\tan u}{1 - \tan^2 u}$$

Pythagorean Identities

$sin^2x + cos^2x = 1$ $\quad tan^2x + 1 = sec^2x$ $\quad 1 + cot^2x = csc^2x$

Power Reducing Identities

$cos^2x = \frac{1}{2}(1 + \cos(2x))$

$sin^2x = \frac{1}{2}(1 - \cos(2x))$

Hyperbolic Trig:

$\sinh(x) = \frac{e^x - e^{-x}}{2}$; $\cosh(x) = \frac{e^x + e^{-x}}{2}$